Complex DSSYK: a normalized route to the first singlet spectral function

Alan Day Chengheng*

Research note — 5 November 2026

Abstract

We connect the Susskind-line singlet programme to existing solutions of complex double-scaled SYK and embedded ensembles. A finite-N dictionary fixes the fermion and coupling normalizations. The first derivative composite is expressed through neighboring particle-number sectors, turning its dynamical spectrum into a precisely specified charged four-point density. Exact ensemble moments and small-system calculations provide the calibration for the next large-N calculation.

1 Fixing the microscopic model

Let N count complex modes, p count creation operators per monomial, and $\lambda = p^2/N$. Work with $\{c_i, c_j^\dagger\} = \delta_{ij}$, $Q = \sum_i c_i^\dagger c_i$, and the full infinite-temperature trace $\mathrm{tr} = 2^{-N}\,\mathrm{Tr}$. The fermions in [1, 3] have $\psi_i = \sqrt{2}c_i$. Thus the string Hamiltonian of [2] becomes

$$H_s = 2^p \sum_{|I|=|J|=p} J_{IJ} c_I^\dagger c_J, \qquad J_{IJ} = J_{JI}^*, \qquad \mathbb{E} J_{IJ} J_{KL} = v_s \delta_{IL}\delta_{JK}, \tag{1}$$

$$v_s = \frac{\mathcal{J}_s^2}{\lambda}\left(\frac{p!}{N^p}\right)^2. \tag{2}$$

Creation products are ordered and annihilation products are their adjoints. Reversing both orderings changes no monomial. Diagonal J_{II} are real with variance v_s; off-diagonal entries have $\mathbb{E}|J_{IJ}|^2 = v_s$. In standard-CAR code the coefficient variance is $4^p v_s$, not v_s.

The BNR convention [3], (2.4), uses $v_B = J_B^2 \binom{N}{p}^{-2}$. The exact coefficient matching is

$$J_B = \frac{\mathcal{J}_s}{\sqrt{\lambda}} P_{N,p}, \qquad P_{N,p} = \frac{(N)_p}{N^p} = \prod_{j=0}^{p-1}\left(1 - \frac{j}{N}\right). \tag{3}$$

At fixed $\lambda > 0$, $P_{N,p} \to e^{-\lambda/2}$, since $\log P_{N,p} = -p(p-1)/(2N) + O(p^3/N^2)$. Keeping this factor is essential when transferring an existing spectral formula.

*AI research assistant (OpenAI).

2 Reuse the known Hamiltonian solution

BNR [3], (4.3), gives the zero-chemical-potential chord moments. After (3), their double-scaled form in our units is

$$\mathbb{E}\,\mathrm{tr}\,H_s^{2k} = \left(\frac{\mathcal{J}_s^2}{\lambda}\right)^k \sum_{\pi\in\mathcal{P}_2(2k)} q^{\mathrm{cr}(\pi)}, \qquad q = e^{-4\lambda}, \qquad \mathbb{E}\,\mathrm{tr}\,H_s^{2k+1} = 0. \tag{4}$$

Here the equality is in the double-scaled limit and $\mathcal{P}_2$ denotes pairings. In particular,

$$\mathbb{E}\,\mathrm{tr}\,H_s^2 = \frac{\mathcal{J}_s^2}{\lambda}, \qquad \frac{\mathbb{E}\,\mathrm{tr}\,H_s^4}{(\mathbb{E}\,\mathrm{tr}\,H_s^2)^2} = 2 + e^{-4\lambda}. \tag{5}$$

The existing q-Hermite measure can therefore be imported directly; there is no need to rederive the Hamiltonian density of states.

Time units. Use $t_s = p\,t_c$ and $H_c = pH_s$, so that $H_s t_s = H_c t_c$. These relations follow the time-unit convention in [4, 5]. For a fixed probe O, $\omega_c = p\omega_s$ and $S_c(\omega_c) = p^{-1}S_s(\omega_c/p)$. A derivative probe also changes: $M_{n,c} = p^n M_{n,s}$, hence

$$A_{2,c} = p^2 A_{2,s}, \qquad S_{22,c}(\omega_c) = p^3 S_{22,s}(\omega_c/p). \tag{6}$$

The coupling symbols in different papers also change definitions. For the same unit Gaussian coefficients, the 2023 complex Hamiltonian and the 2025 one obey $H_{2025} = \mathcal{J}_{2025}H_{2023}/(p\mathcal{J}_{2023})$. We fix (2) first and perform time conversion only once. The Majorana paper [4] instead uses total degree p_M and $\lambda_M = 2p_M^2/N_M$; its coefficients are not those of the complex model.

Finite-N normalization. Set $w = 4^p v_s$. In the m-particle sector, of dimension $d_m = \binom{N}{m}$, the standard embedded-ensemble identity [6], (2.8)–(2.9), gives

$$\mathbb{E}(H_s^2)|_m = F_m I_m, \qquad F_m = w\binom{m}{p}\binom{N-m+p}{p}, \qquad F_m = 0 \ (m < p). \tag{7}$$

The full trace second moment is exactly $2^{-N}\sum_m d_m F_m$. Equivalently, counting the overlap $k = |I \cap J|$,

$$\mathbb{E}\,\mathrm{tr}\,H_s^2 = v_s\binom{N}{p}\sum_{k=\max(0,2p-N)}^{p}\binom{p}{k}\binom{N-p}{p-k}2^k. \tag{8}$$

Equations (7) and (8) are independent implementations of the same finite-system normalization, used below as a check.

3 The first singlet connects neighboring charge sectors

Use $O(t) = e^{iH_s t} O e^{-iH_s t}$ and $M_n = \sum_i c_i^\dagger (i\,\mathrm{ad}_{H_s})^n c_i$. As in the programme note, $M_0 = Q$, $M_1 = -ipH_s$ and

$$A_2 = -M_2 = \sum_i [H_s, c_i]^\dagger [H_s, c_i]. \tag{9}$$

Define $\mathcal{T}(X) = \sum_i c_i^\dagger X c_i$. The homogeneous normal-ordered CAR identity gives $\mathcal{T}(H_s) = (Q-p)H_s$. Expanding the two commutators then yields the useful exact formula

$$\boxed{A_2 = \mathcal{T}(H_s^2) - (Q-2p)H_s^2.} \tag{10}$$

Let H_m denote H_s restricted to particle number m. Since c_i lowers particle number by one,

$$(A_2)_m = \sum_i c_i^\dagger H_{m-1}^2 c_i - (m-2p)H_m^2. \tag{11}$$

The second term is conserved. All nonzero-frequency weight comes from the first term, which retains the same couplings in adjacent sectors. For energy eigenstates r, s in sector m and u in sector $m-1$,

$$(A_2)_{rs}^{(m)} = \sum_{i,u} (E_u^{(m-1)})^2 (c_i)_{ur}^* (c_i)_{us} - (m-2p)(E_r^{(m)})^2 \delta_{rs}. \tag{12}$$

This identifies the matrix elements needed for the singlet spectrum. The coherent sum over i, u must be formed before taking its modulus squared. The Hamiltonian density of states alone does not supply it.

Two useful exact consequences. At the first active sector, $m = p$, the preceding Hamiltonian vanishes, so $(A_2)_p = pH_p^2$. This sector has only zero-frequency weight. For $p = 1$, $H = c^\dagger h c$ and $A_2 = c^\dagger h^2 c$ commute in every sector. These give two distinct conserved controls for a calculation.

Taking the ensemble expectation in (11) and using (7) also fixes the mean without numerical disorder sampling:

$$\begin{aligned} \mathbb{E}(A_2)|_m &= \left[mF_{m-1} - (m-2p)F_m\right] I_m \\ &= F_m \frac{p(N-p+1)}{N-m+1} I_m, \qquad m \geq p. \end{aligned} \tag{13}$$

The elementary binomial ratio proves the second line. This is a normalization and static-background formula, not the dynamical spectrum.

4 The precise four-point object to compute next

Write $D_m(E) = \delta(H_m - E)$, understood as a finite-system spectral measure. Define the charged four-point density

$$\Xi_m(E, E'; u, v) = \frac{1}{d_m}\mathbb{E}\sum_{i,j} \mathrm{Tr}_m\,[c_i^\dagger D_{m-1}(u) c_i D_m(E') \cdot c_j^\dagger D_{m-1}(v) c_j D_m(E)]. \tag{14}$$

In this equation u, v are energy variables, not state labels. The first probe's nonzero-frequency trace spectrum is exactly

$$S_{22}(\omega) = 2\pi \sum_{m=1}^{N} \frac{d_m}{2^N} \int dE\, dE'\, du\, dv\; u^2 v^2 \cdot \Xi_m(E, E'; u, v)\delta(\omega - E' + E), \qquad \omega \neq 0. \tag{15}$$

Centering A_2 changes only its zero-frequency contribution. Likewise, the conserved term in (11) drops out of (15). The formula follows by inserting energy projectors in (12); it is an exact finite-N reformulation of the proposed observable.

Why this is an efficient next target. BNR supplies oriented-chord methods for complex fermions, while Tall–Tomsovic supplies fixed-charge moments and four-point strength densities [3, 6]. We can use those methods to evaluate (14), retaining both adjacent sectors and the prescribed index contractions. This is more specific than constructing an unspecified new ladder kernel from scratch.

The published closed probe formulas in BNR §7–8 and Tall–Tomsovic §4 use independent Gaussian probe coefficients. Our A_2 depends quadratically on the same J as H_s. Equation (14) keeps that dependence through explicit elementary fermion insertions, rather than replacing A_2 by an independent random operator. Individual elementary insertions have fixed length; their double-scaled limit and the weighted energy integrals must be controlled together.

The calculation sequence. First derive low energy-weighted moments of (14) using the fixed-charge operator identities and oriented contractions. Reproduce the conserved channels and (13). Then resum the dynamical channel, starting with small positive λ after double scaling. Finally compare the resulting boost-frequency kernel with Rindler bulk propagation to extract invariant particle masses and weights.

5 Finite-system calibration

We implemented (2) with $\mathcal{J}_s = 1$ and the exact full trace. Five covariance sums over all pairs of Hamiltonian monomials agree with (8) and the mean of A_2; an independent particle-sector sum using (7)–(13) agrees as well. These ensemble checks require no Monte Carlo estimate.

Eighteen same-sample calculations use three seeds (1105–1107) for each pair (N, p) below. Diagonalizing within exact charge sectors, we store every transition frequency and weight, plus the couplings, Hamiltonian and A_2 matrices. Let $O_2 = A_2 - \mathrm{tr}(A_2)I$ and

$$G = \mathrm{tr}\, O_2^2, \qquad K = \mathrm{tr}([H_s, O_2]^\dagger [H_s, O_2]), \qquad f_{\mathrm{dyn}} = \frac{\int_{\omega\neq 0} S_{22}(\omega) d\omega}{2\pi G}. \tag{16}$$

The observed ranges of f_{dyn} are

N	p	f_{dyn} across three samples
4	1	below 4×10^{-28} (conserved)
3	2	below 3×10^{-31} (conserved)
4	2	0.02414–0.07011
5	2	0.05992–0.10605
6	2	0.12430–0.15191
6	3	0.03598–0.03757

Thus the construction distinguishes the conserved controls from interacting samples with nonzero dynamical weight. These small systems calibrate the observable; they are not an extrapolation of its double-scaled spectrum. The 178 algebraic and spectral checks include M_1, M_2, (11), the $m = p$ sector and both spectral sum rules. The largest recorded scaled residual is 1.30×10^{-15}.

A Reproduction and remaining scope

The implementation and read-only audit scripts are listed in the accompanying source ledger and stored in `code/`. The environment is Python 3.14.5 and NumPy 2.4.6, also recorded in `calibration.json`. All calculations use complex double precision and save code/data hashes. The numerical zero bin is $10^{-10} \max(1, \|H_s\|_2)$; reducing it by ten changes no reported static weight at the test tolerance.

The algebraic formulas are standard CAR/ensemble consequences; the Hamiltonian chord solution is due to the cited literature. The large-N evaluation of (14), its spectral normalization and its bulk mass map remain the next research work. Late-time exponential approximations to elementary correlators are not differentiated at zero time to obtain the energy-weighted insertions above.

References

[1] L. Susskind, *De Sitter Space has no Chords. Almost Everything is Confined*, arXiv:2303.00792v1.

[2] S. Miyashita, Y. Sekino and L. Susskind, *DSSYK at Infinite Temperature: The Flat-Space Limit and the 't Hooft Model*, arXiv:2506.18054v3.

[3] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev-Ye-Kitaev model in the double scaling limit*, JHEP 02 (2021) 113, arXiv:2006.13983v2.

[4] Y. Sekino and L. Susskind, *Double-Scaled SYK, QCD, and the Flat Space Limit of de Sitter Space*, arXiv:2501.09423v2.

[5] A. Rahman and L. Susskind, *Infinite Temperature is Not So Infinite: The Many Temperatures of de Sitter Space*, arXiv:2401.08555v3.

[6] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2.